

Explainable Network Verification via Subspecifications

User Study - Introduction of Background Knowledge

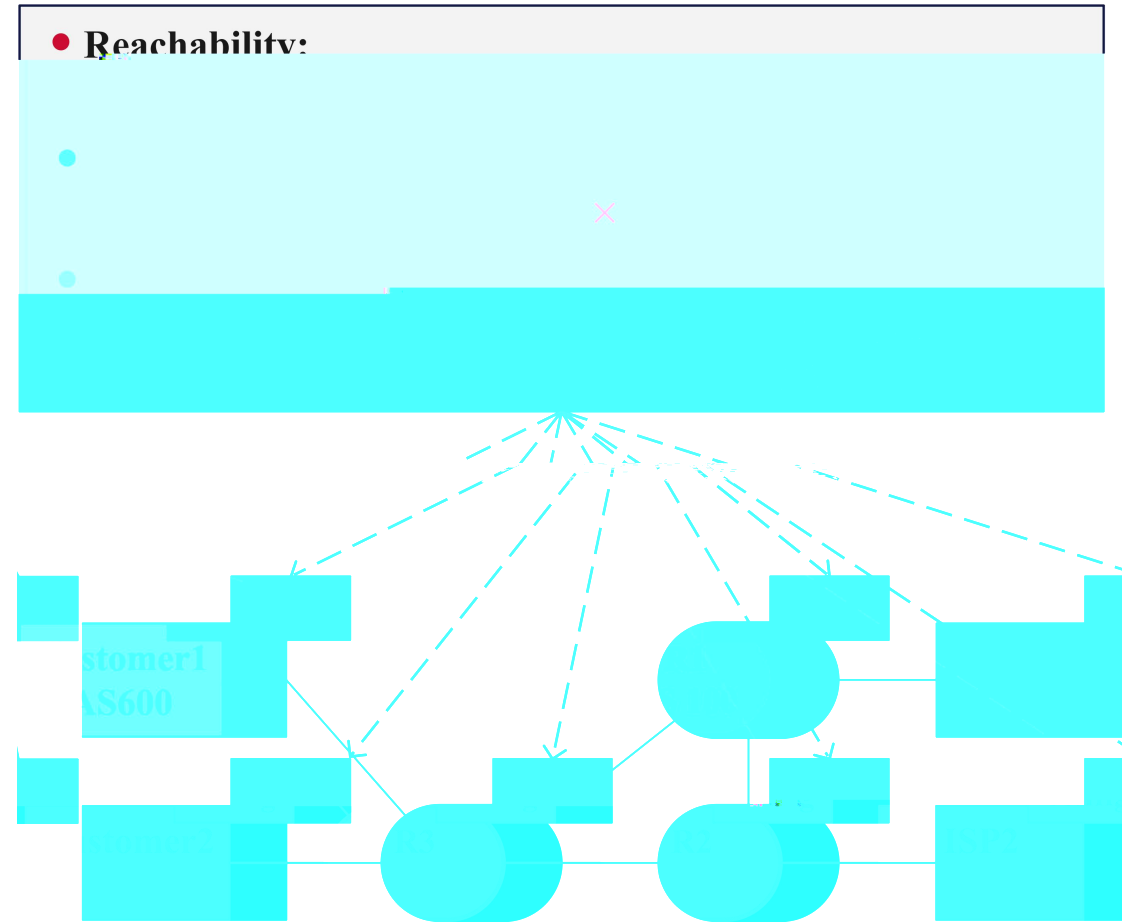
Network Operations and Network Verifications



For network operators, **aligning** network specifications (network routing policies or network design intents) with the actual device configurations is **an extremely challenging task**.



Network verifier ensures that device configurations **faithfully implement** the network specifications (network routing policies or network design intents) by using formal methods, providing **mathematically correctness guarantees**.



Background: Explainable Network Verification

Network verifiers often give **YES** **NO** (*with a counterexample*) answers, without explaining *why*.

Example: suppose we want a BGP policy blocking the private prefix 192.168.0.0/16.

```
1 ip prefix-list a seq 5 permit 192.168.0.0/ ge
2 !
3 ip prefix-list a seq 5 permit 192.168.0.0/ ge
```

Both of them pass the verification of the network verifier.

However, **a** is more **precise** than **a** .

Overly restrictive filter may block more prefix than necessary.

Explainable Network Verification

Why does a specific configuration field or line satisfy **the network specification**
(which blocks the private prefix 192.168.0.0/16)?

permit **AND** prefix cover 192.168.0.0/16 **AND**
prefix ge range = 0~16 **AND**
prefix ge range >= prefix length

permit

prefix cover 192.168.0.0/16

1		a	seq 5			
2	!					
3	ip prefix-list	a	seq 5	permit	192.168.0.0/16	ge 16

prefix ge range =
12~16

Both of them pass the verification of the network verifier.

Explainable Network Verification

Why does a specific configuration field or line satisfy **the network specification** (which blocks the private prefix 192.168.0.0/16)?

permit AND prefix cover 192.168.0.0/16 AND
prefix ge range = 0~16 AND
prefix ge range >= prefix length

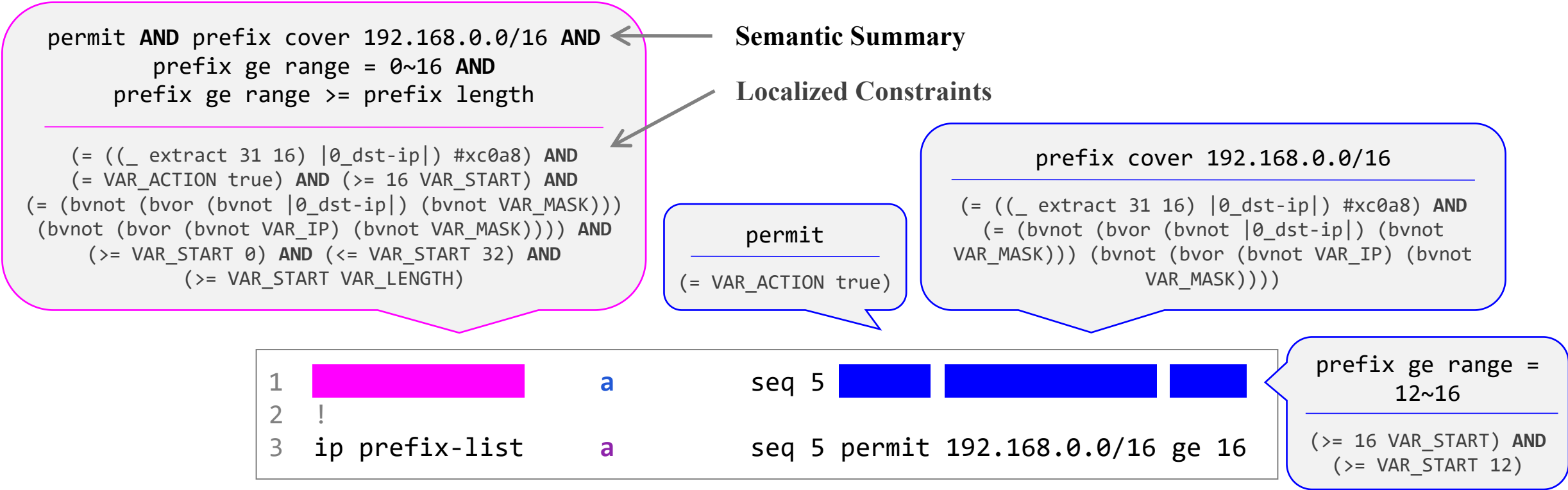
1		a	seq 5 permit	192.168.0.0/12	ge 12
2	!				
3	ip prefix-list	a	seq 5 permit	192.168.0.0/16	ge 16

Both of them pass the verification of the network verifier.

a is the more precise option.

Explainable Network Verification via Subspecifications

Localized Subspecifications (Subspecs): the safe modification scope of a configuration field or line that preserves the prior verification success of the network specification (which blocks the private prefix 192.168.0.0/16).



Case 1: Modify a **Single** Configuration Field

The previously verified network specification: blocking the private prefix.

- Modification **within** subspec: preserves the verified specification.

prefix ge range = 12~16
(>= 16 VAR_START) AND
(>= VAR_START 12)

1	ip prefix-list	a	seq	permit	192.168.0.0/12		
1	ip prefix-list	a	seq	permit	192.168.0.0/12		



- Modification **out of** subspec: may violate the verified specification. *(sound but not complete)*

prefix cover 192.168.0.0/16
(= ((_ extract 31 16) |0_dst-ip|) #xc0a8) AND
(= (bvnot (bvor (bvnot |0_dst-ip|) (bvnot VAR_MASK)))
(bvnot (bvor (bvnot VAR_IP) (bvnot VAR_MASK))))

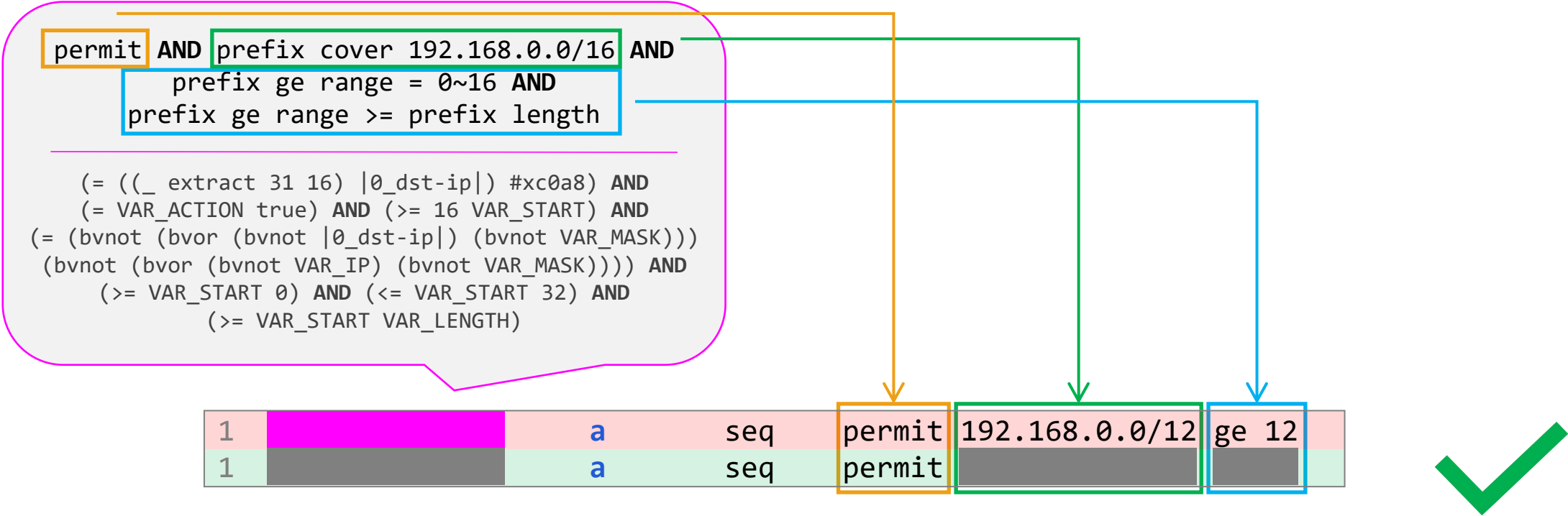
1	ip prefix-list	a	seq	permit		ge 12	
1	ip prefix-list	a	seq	permit		ge 12	



Case 2: Modify **Multiple** Configuration Fields

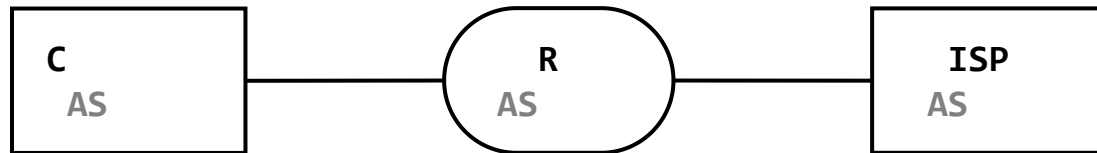
The previously verified network specification: blocking the private prefix.

- Modification **multiple configuration fields** **within** subspec: preserves the verified specification.



Tips for User Study

- In this user study, the eBGP best route selection process only involves **AS-path length**.
- In this user study, the route-map naming rule is **Router_Direction(IN_FROM/OUT_TO)_Peer**.
 - R1_IN_FROM_ISP1 — BGP import policy on R1 from ISP1 ($R1 \leftarrow ISP1$).
 - R1_OUT_TO_Customer — BGP export policy on R1 to Customer ($R1 \rightarrow Customer$).



Explainable Network Verification via Subspecifications

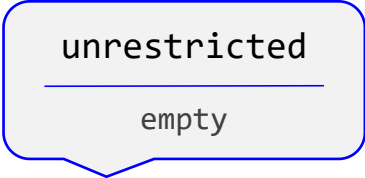
User Study - Introduction of Background Knowledge

Thank you for participating in this user study!

Case 1: Modify a **Single** Configuration Field (Empty Subspec)

The previously verified network specification: blocking the private prefix.

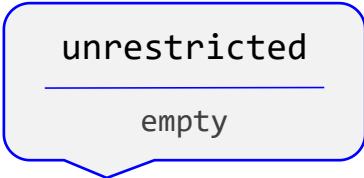
■ **Arbitrary modification:** preserves the verified specification.



1	ip prefix-list	a	seq	permit	192.168.0.0/12		
1	ip prefix-list	a	seq	permit	192.168.0.0/12		
2	ip prefix-list	a	seq	permit	10.0.0.0/8	ge 8	



■ **Arbitrary modification:** preserves the verified specification.



1	ip prefix-list	a	seq	permit		ge 12	
1	ip prefix-list	a	seq	permit		ge 12	
2	ip prefix-list	a	seq	permit	10.0.0.0/8	ge 8	



Case 2: Modify **Multiple** Configuration Fields (Empty Line-level Subspec)

The previously verified network specification: blocking the private prefix.

- **Arbitrary modification** multiple configuration fields: preserves the verified specification.

unrestricted

empty

1		a	seq	permit	192.168.0.0/12	ge 12
1		a	seq	d		
2	ip prefix-list	a	seq	permit	10.0.0.0/8	ge 8



- **Safe removal** the entrie configuration line: preserves the verified specification.

unrestricted

empty

1		a	seq	permit	192.168.0.0/12	ge 12
1		private_ips	seq	permit	192.168.0.0/12	ge 12
2	ip prefix-list	a	seq	permit	10.0.0.0/8	ge 8

